

Figure 4: Basic principle of SMOTE algorithm (Image credits: <https://rikunert.com/>)

- The source code is:

```
from collections import Counter
from sklearn.datasets import make_classification
from imblearn.over_sampling import SMOTE
from matplotlib import pyplot
from numpy import where

X, y = make_classification(n_classes=2, class_sep=2,
weights=[0.1, 0.9], n_features=4, n_clusters_per_class=1,
n_samples=1000, random_state=10)

# summarize class distribution
print('Original dataset shape %s' % Counter(y))
Original dataset shape Counter({1: 894, 0: 106})

# perform upsampling
sm = SMOTE(random_state=42)
X_res, y_res = sm.fit_resample(X, y)
print('Resampled dataset shape %s' % Counter(y_res))
Resampled dataset shape Counter({1: 894, 0: 894})

# scatter plot of examples by class label post upsampling
for label, _ in Counter(y_res).items():
    row_ix = where(y_res == label)[0]
    pyplot.scatter(X_res[row_ix, 0], X_res[row_ix, 1],
label=str(label))
pyplot.legend()
pyplot.show()
```

ADASYN (Adaptive Synthetic)

- Oversampling can be done using the Adaptive Synthetic (ADASYN) algorithm. This algorithm works similarly to SMOTE (can be considered as an extension), but it generates a varying number of samples based on the local distribution of the class that needs to be oversampled.
- **Logic**
 - ADASYN is a technique that generates synthetic samples for minority classes using the feature space of the original dataset. It calculates the density distribution of each minority class sample and

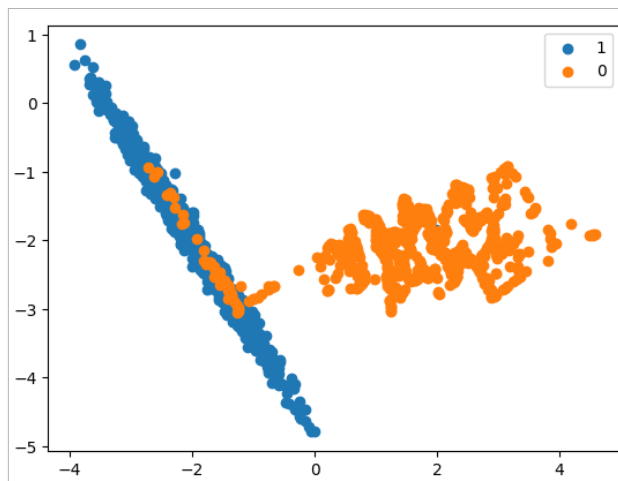


Figure 5: Scatter plot of examples by class label upsampled post using SMOTE

generates synthetic samples according to the density distribution. The main idea behind ADASYN is to use a weighted distribution for different minority class examples based on their level of difficulty in learning. It generates more synthetic data for minority class examples that are harder to learn compared to those that are easier to learn.

- Randomly selects an instance from the minority class and identifies its k nearest minority class neighbours (usually $k=5$).
- Finds the ratio $r_i = \text{majority_class}$ in neighbourhood / K .
- The r_i value indicates the dominance of the majority class in each specific neighbourhood. Higher r_i neighbourhoods contain more majority-class examples and are more difficult to learn.
- Because r_i is higher for neighbourhoods dominated by majority class examples, more synthetic minority class examples will be generated for those neighbourhoods. This is what gives the ADASYN algorithm its adaptive nature, generating more data for harder-to-learn neighbourhoods.
- The main difference between ADASYN and SMOTE is that ADASYN generates synthetic samples adaptively based on the density distribution of minority class samples, while SMOTE generates synthetic samples by interpolating between minority class samples. This adaptive approach helps to focus more on difficult-to-learn samples, potentially leading to better classification performance.
- However, ADASYN has a limitation of increased computational complexity due to the generation of synthetic samples. This may affect the training time of machine learning models.